INCH-POUND

MIL-M-38510/344
AMENDMENT 3
17 June 1993
SUPERSEDING
AMENDMENT 2
16 November 1992

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED
SCHOTTKY TTL, DECADE COUNTERS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/344, dated 12 June 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3, thermal resistance: Delete and substitute as follows:

"Thermal resistance, junction-to-case (Θ_{JC}):
Cases E, F, and 2 - - - - - - - - - - - - - - - - - - (See MIL-M-38510, appendix C)".

1.3, after junction temperature (T_J), delete "3/" and substitute "2/".

Footnotes 2/ and 3/: Delete in their entirety and substitute as follows:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 4

TABLE I, symbol column, below I_{IL2}: Add "I_{IL3}".

PAGE 6

TABLE I, t_{PHL6}, device type 04, minimum limits column: Delete "7.0" and substitute "6.0".

TABLE I, t_{PHL11}, device type 04, minimum limits column: Delete "6.5" and substitute "5.5".

TABLE I, t_{PLH9}, device type 04, minimum limits column: Delete "7.0" and substitute "6.0".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
23	23
24	Reprint without change

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

PAGE 7

TABLE I, t_{PHL9}, device type 04, minimum limits column: Delete "6.5" and substitute "5.5".

TABLE I, t_{PLH10}, device type 04, minimum limits column: Delete "7.0" and substitute "6.0".

TABLE I, t_{PHL10}, device type 04, minimum limits column: Delete "6.5" and substitute "5.5".

PAGE 18

Note 2, delete and substitute:

"2. Inputs not under test are at ground or at 2.7 V as specified in table III."

PAGE 19

FIGURE 5, directly below clock: Delete "t_{PLH}" and substitute "t_{PLH1}".

PAGE 27

TABLE III, $\overline{PE}$ pin, test numbers 5 through 10: Add "„".

TABLE III, $\overline{PE}$ pin, test numbers 47 through 51: Add "0.0 V".

TABLE III, top of page: Delete "Terminal conditions (pins not designated may be high 2.0 V, low 0.8 V, or open)" and substitute "Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)". Make this same change on pages 28 through 39.

PAGE 28

TABLE III, Q_0 pin, test number 114: Delete "„" and substitute "H".

PAGE 29

* TABLE III, device type 01, symbol column: Add "9/" after f_{MAX}.

TABLE III, $\overline{PE}$ pin, test numbers 129 through 136: Add "0.0 V".

TABLE III, P_0 pin, test number 133: Add "0.0 V".

TABLE III, P_1 pin, test number 134: Add "0.0 V".

TABLE III, P_2 pin, test number 135: Add "0.0 V".

TABLE III, P_3 pin, test number 136: Add "0.0 V".

TABLE III, P_3 pin, test number 140: Add "2.7 V".

TABLE III, $\overline{PE}$ pin, test numbers 141 through 145: Add "0.0 V".

PAGE 30

TABLE III, $\overline{PE}$ pin, test number 5: Add "0.8 V".

TABLE III, $\overline{PE}$ pin, test numbers 47 through 51: Add "0.0 V".

PAGE 31

TABLE III, TC pin, test number 75: Delete "L" and substitute "H".

TABLE III, TC pin, test number 77: Delete "H" and substitute "L".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

TABLE III, TC pin, test number 78: Delete "„" and substitute "H".

TABLE III, TC pin, test number 87: Delete "H" and substitute "L".

PAGE 32

TABLE III, Q_0 pin, test number 125: Delete "„" and substitute "H".

* TABLE III, device type 02, symbol column: Add "9/" after f_{MAX}.

TABLE III, P_0 pin, test number 144: Add "0.0 V".

TABLE III, P_1 pin, test number 145: Add "0.0 V".

TABLE III, P_2 pin, test number 146: Add "0.0 V".

TABLE III, P_3 pin, test number 147: Add "0.0 V".

PAGE 33

TABLE III, measured terminal column, test number 19: Delete "$\overline{RC}$" and substitute "CP".

TABLE III, measured terminal column, test number 20: Delete "CP" and substitute "P_0".

TABLE III, measured terminal column, test number 27: Delete "$\overline{RC}$" and substitute "CP".

TABLE III, measured terminal column, test number 28: Delete "CP" and substitute "P_0".

TABLE III, measured terminal column, test number 35: Delete "$\overline{RC}$" and substitute "CP".

TABLE III, measured terminal column, test number 36: Delete "CP" and substitute "P_0".

TABLE III, symbol column, test number 44: Delete "I_{IL2}" and substitute "I_{IL3}".

TABLE III, $\overline{PL}$ pin, test numbers 45 through 50: Add "0.0 V".

TABLE III, P_1, $\overline{CE}$, $\overline{U/D}$, and P_2 pins, test numbers 49 and 50: Add "0.0 V".

PAGE 34

TABLE III, P_3 and P_2 pins, test number 57: Delete "B" and substitute "A".

TABLE III, P_1, P_3, P_2, and P_0 pins, test number 91: Add "0.0 V".

TABLE III, P_1, P_3, and P_2 pins, test number 92: Add "0.0 V".

TABLE III, P_0 pin, test number 92: Add "2.7 V".

TABLE III, P_3 and P_2 pins, test number 93: Add "0.0 V".

TABLE III, P_1 and P_0 pins, test number 93: Add "2.7 V".

TABLE III, P_3 pin, test number 94: Add "0.0 V".

TABLE III, P_1, P_2, and P_0 pins, test number 94: Add "2.7 V".

TABLE III, P_1, P_3, and P_2 pins, test number 95: Add "0.0 V".

TABLE III, P_0 pin, test number 95: Add "2.7 V".

TABLE III, P_3 and P_2 pins, test number 96: Add "0.0 V".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

TABLE III, P_1 and P_0 pins, test number 96: Add "2.7 V".

TABLE III, P_3 pin, test number 97: Add "0.0 V".

TABLE III, P_1, P_2, and P_0 pins, test number 97: Add "2.7 V".

TABLE III, P_1 and P_2 pins, test number 98: Add "0.0 V".

TABLE III, P_3 and P_0 pins, test number 98: Add "2.7 V".

TABLE III, P_1, P_3, P_2, and P_0 pins, test number 99: Add "0.0 V".

TABLE III, $\overline{CE}$ and $\overline{U}/D$ pins, test numbers 91 through 102: Add "$_{\text{H}}$".

TABLE III, P_1 and P_2 pins, test numbers 100 through 102: Add "0.0 V".

TABLE III, P_0 pin, test numbers 100 through 102: Add "2.7 V".

PAGE 35

TABLE III, $\overline{PL}$ pin, test numbers 103 through 110: Delete "2/" and substitute "0.0 V".

TABLE III, P_1, $\overline{U}/D$, P_2, and CP pins, test numbers 111 and 112: Add "0.0 V".

TABLE III, P_0 pin, test number 117: Add "0.0 V".

TABLE III, P_1 pin, test number 118: Add "0.0 V".

TABLE III, P_2 pin, test number 119: Add "0.0 V".

TABLE III, P_3 pin, test number 120: Add "0.0 V".

TABLE III, P_1, $\overline{CE}$, and P_2 pins, test numbers 121 and 122: Add "0.0 V".

TABLE III, P_1, $\overline{CE}$, P_3, P_2, and P_0 pins, test numbers 123 and 124: Add "0.0 V".

PAGE 36

TABLE III, P_3 pin, test number 4: Add "2.0 V".

TABLE III, P_1 pin, test number 5: Delete "5.5 V" and substitute "0.0 V".

TABLE III, P_2 and $\overline{PL}$ pins, test number 5: Add "0.0 V".

TABLE III, CPU pin, test number 6: Add "5.5 V".

TABLE III, P_1, $\overline{PL}$, and P_0 pins, test number 6: Add "0.0 V".

TABLE III, P_3 and P_2 pins, test number 6: Delete "5.5 V" and substitute "0.0 V".

TABLE III, CPD pin, test number 11: Add "5.5 V".

TABLE III, P_1 and P_2 pins, test number 11: Delete "5.5 V" and substitute "0.0 V".

TABLE III, CPU pin, test number 12: Add "5.5 V".

TABLE III, P_1, P_3, P_2, and P_0 pins, test number 12: Delete "0.8 V" and substitute "0.0 V".

TABLE III, P_3 pin, test number 38: Add "0.5 V".

TABLE III, P_3 pin, test number 39: Delete "0.5 V".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

TABLE III, P_2 pin, test number 39: Add "0.5 V".

TABLE III, P_2 pin, test number 40: Delete "0.5 V".

TABLE III, symbol column, test numbers 43 and 44: Delete "I_{IL1}" and "I_{IL2}" respectively and substitute "I_{IL3}".

TABLE III, $\overline{PL}$ pin, test number 49: Add "0.0 V".

TABLE III, P_1, P_3, P_2, $\overline{PL}$, MR, and P_0 pins, test number 50: Add "0.0 V".

PAGE 37

TABLE III, test numbers 77 and 78, pin Q_3: Delete "H" and substitute "L".

TABLE III, MR pin, test number 96: Delete "A" and substitute "B".

TABLE III, P_0 pin, test number 96: Delete "H" and substitute "A".

PAGE 38

* TABLE III, device type 04, symbol column: Add "9/" after f_{MAX}.

TABLE III, P_0 pin, test number 140: Add "2.7 V".

TABLE III, P_1 and P_0 pins, test number 141: Add "2.7 V".

TABLE III, P_1, P_2, and P_0 pins, test number 142: Add "2.7 V".

TABLE III, P_1 pin, test number 143: Add "2.7 V".

TABLE III, P_2 pin, test number 144: Add "2.7 V".

TABLE III, P_3 pin, test number 145: Add "2.7 V".

TABLE III, P_0 pin, test number 147: Add "2.7 V".

TABLE III, P_1 and P_0 pins, test number 148: Add "2.7 V".

TABLE III, P_1, P_2, and P_0 pins, test number 149: Add "2.7 V".

TABLE III, P_1, P_3, P_2, and P_0 pins, test number 150: Add "2.7 V".

TABLE III, P_0 pin, test number 151: Add "2.7 V".

TABLE III, P_1 pin, test number 152: Add "2.7 V".

TABLE III, P_2 pin, test number 153: Add "2.7 V".

TABLE III, CPU and P_3 pins, test number 154: Add "2.7 V".

TABLE III, P_3 pin, test number 155: Add "2.7 V".

TABLE III, P_3 and P_0 pins, test number 156: Add "2.7 V".

TABLE III, P_0 pin, test number 166: Add "2.7 V".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

PAGE 39

TABLE III, P_3 and P_0 pins, test numbers 175 and 176: Add "2.7 V".

TABLE III, t_{PHL6}, test number 176, minimum limits column: Delete "7.0" and substitute "6.0".

TABLE III, P_3 pin, test number 177: Add "2.7 V".

TABLE III, t_{PLH7}, t_{PLH8}, t_{PHL7}, and t_{PHL8}, test numbers 177 through 180, minimum limits column: Delete "„" and substitute "7.0".

TABLE III, P_3 and P_0 pins, test numbers 178 and 179: Add "2.7 V".

TABLE III, P_3 pin, test number 181: Add "2.7 V".

TABLE III, t_{PLH9}, test numbers 181 and 182, minimum limits column: Delete "„" and substitute "6.0".

TABLE III, P_0 pin, test number 182: Add "2.7 V".

TABLE III, $\overline{TCU}$ pin, test number 182: Add "„".

TABLE III, P_3 pin, test number 183: Add "2.7 V".

TABLE III, t_{PHL9}, test numbers 183 and 184, minimum limits column: Delete "6.5" and substitute "5.5".

TABLE III, P_0 pin, test number 184: Add "2.7 V".

TABLE III, $\overline{TCU}$ pin, test number 184: Add "„".

TABLE III, t_{PLH10}, test numbers 185 through 188, minimum limits column: Delete "7.0" and substitute "6.0".

TABLE III, t_{PHL10}, test numbers 189 through 192, minimum limits column: Delete "6.5" and substitute "5.5".

TABLE III, P_0 pin, test number 193: Add "2.7 V".

TABLE III, t_{PHL11}, test numbers 193 through 196, minimum limits column: Delete "„" and substitute "5.5".

TABLE III, P_1 pin, test number 194: Add "2.7 V".

TABLE III, P_2 pin, test number 195: Add "2.7 V".

TABLE III, P_3 pin, test number 196: Add "2.7 V".

PAGE 40

* TABLE III, add footnote 9/ as follows:

"9/ f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1348)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.

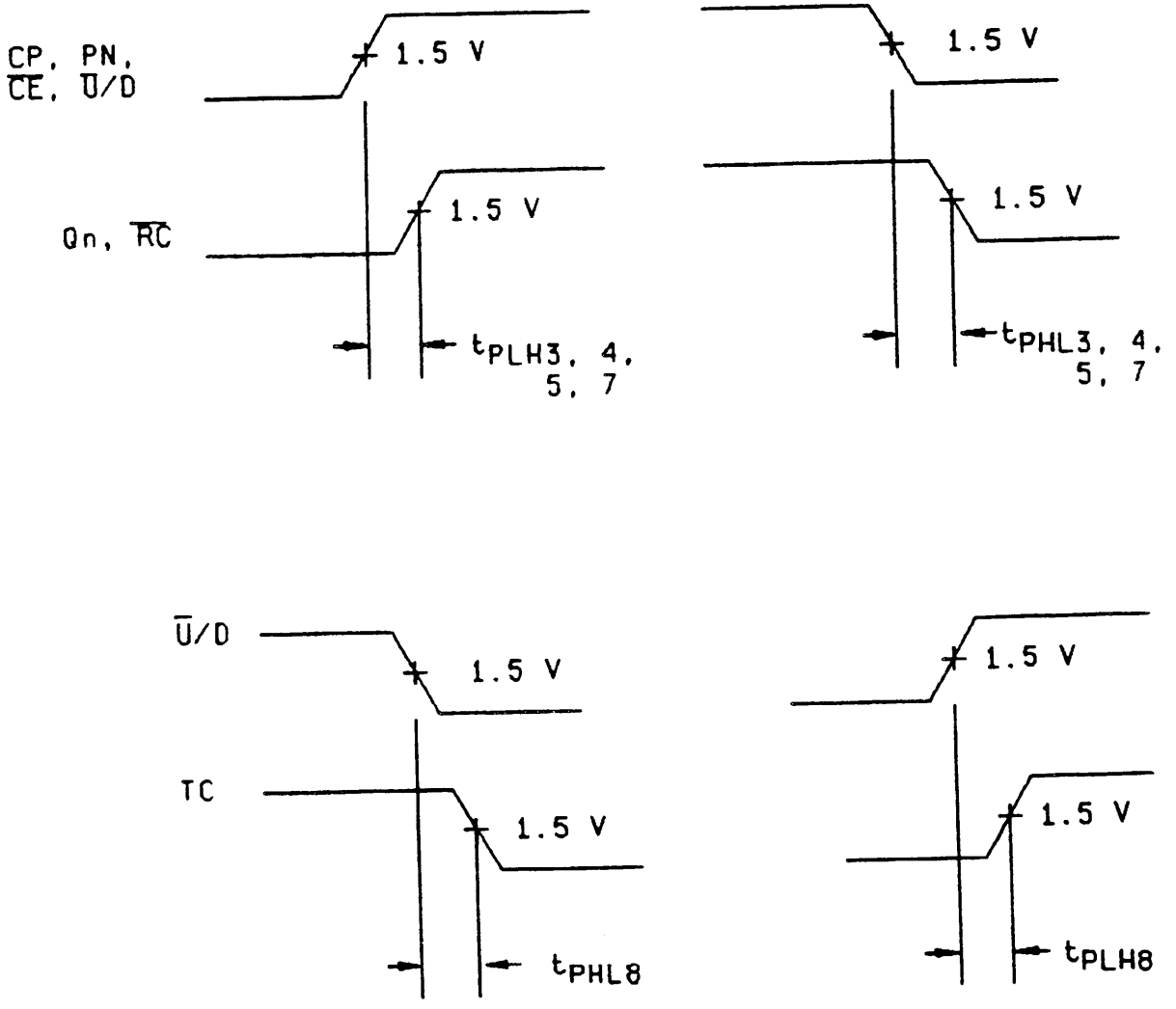
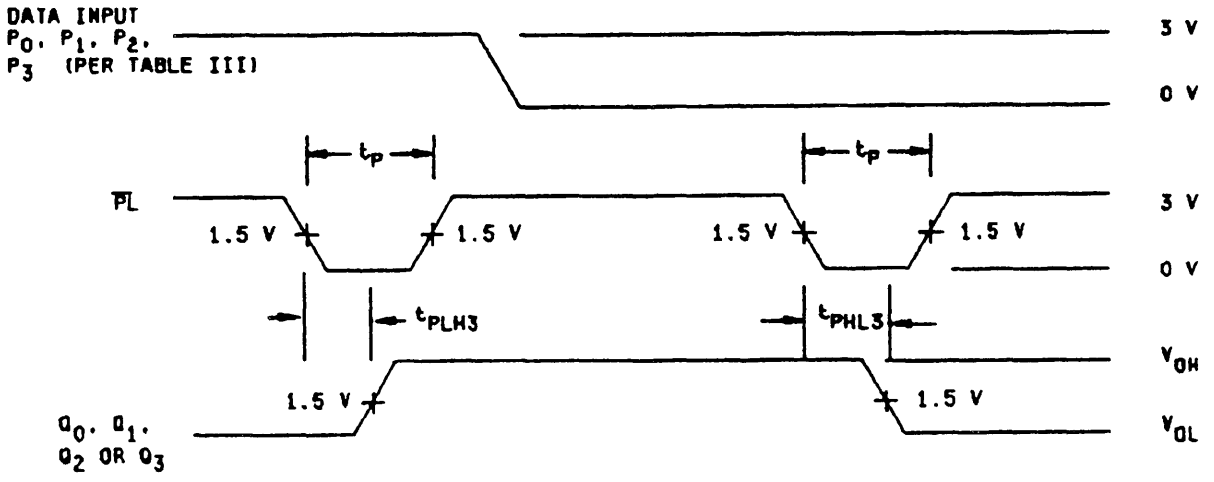
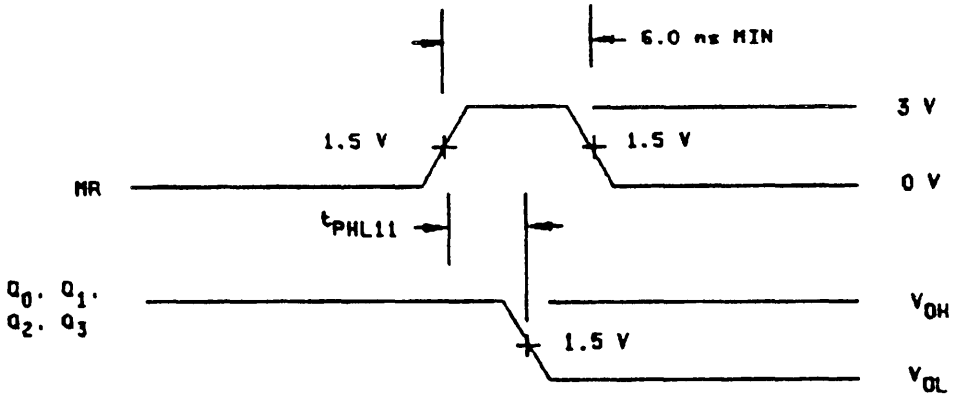


FIGURE 5. Switching time test circuit and waveforms for device type 03 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.



PARALLEL LOADED VOLTAGE WAVEFORMS



CLEAR SWITCHING VOLTAGE WAVEFORMS

FIGURE 5. Switching time circuit and waveforms for device type 04 - Continued.

Reprinted without change

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:22Z
Check the source to verify that this is the current version before use.